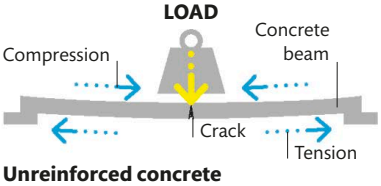
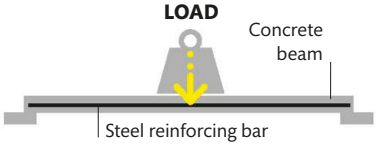
Strengthening concrete

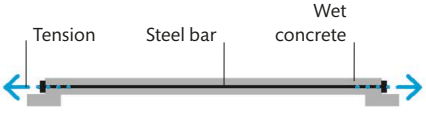
Large concrete structures often use concrete reinforced with steel mesh or bars (called rebars) to increase the strength of the concrete. Concrete can be made even stronger by prestressing—putting the rebars under tension while the concrete is hardening.



Unreinforced concrete
Concrete is strong under compression but relatively weak under tension. A heavy load can cause concrete to bend and crack.



Steel-reinforced concrete
Placing a steel bar inside concrete helps to prevent it from bending and cracking under heavy loads.



Forming prestressed concrete
Concrete is poured around a steel bar that is under tension. As the concrete sets, it bonds to the bar.



Hardened prestressed concrete
When the concrete has set, tension on the bar is released. The bar compresses the concrete, making it stronger.

WHAT IS CONCRETE CANCER?

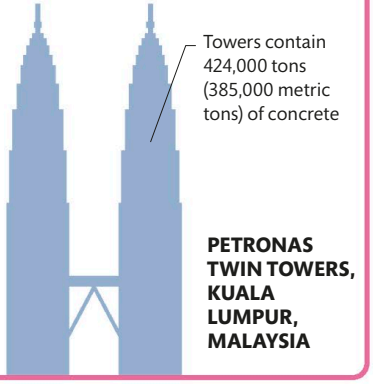
Concrete cancer is staining, cracking, and eventual breaking of reinforced concrete when rust expands the steel inside the concrete, destroying the concrete from within.

THE **ANCIENT ROMANS** USED **VOLCANIC ASH, CALLED POZZOLANA,** TO MAKE **CONCRETE**



MASSIVE CONCRETE STRUCTURES

Many of the world's largest man-made structures are made of concrete. The most massive is the Three Gorges Dam in China, made of more than 72 million tons (65 million metric tons) of concrete. The Petronas Twin Towers is the most massive concrete building.



PETRONAS TWIN TOWERS, KUALA LUMPUR, MALAYSIA

TYPES OF CONCRETE

Type	Characteristics
Precast concrete	Unlike standard concrete, which is cast and cured on-site, precast concrete is cast and cured elsewhere then transported to the construction site and lifted into place.
Heavyweight concrete	Using special aggregates, such as iron, lead, or barium sulphate, heavyweight concrete is much denser than normal concrete and is mainly used for radiation shielding.
Shotcrete	Shotcrete is concrete that is applied by high-pressure spraying, usually onto a steel mesh framework. It is often used for artificial rock walls, tunnel linings, and pools.
Pervious concrete	Pervious concrete is made with coarse particles of aggregate, which makes the concrete porous, allowing water to drain through it.
Rapid-strength concrete	This type of concrete contains additives, such as calcium chloride, to speed up curing so that the concrete becomes strong and hard enough to bear loads within a few hours.
Glass concrete	Glass concrete uses recycled glass as the aggregate. It is stronger and provides better thermal insulation than standard concrete and resembles marble in appearance.